Interest-Driven Personalized Sports Recommendation Using Classical Ensemble Learning with a Novel-Sport Discovery Mechanism

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***Abstract*—Existing sports matching systems rely on physical fitness metrics, leaving personal interests underutilized despite evidence that intrinsic motivation best predicts sustained engagement. We present an *interest-driven* recommendation framework that positions personal interests as the primary signal, with physical and demographic features as secondary factors. Our system uses classical machine learning—hyperparameter-tuned gradient boosting, random forests, and a stacking ensemble with logistic regression as meta-learner—to predict playing and watching preferences. We additionally introduce a *discovery score* based on cosine similarity that surfaces unfamiliar sports aligned with latent interests. Evaluated on a literature-calibrated synthetic dataset of 1,000 users across 20 sports, the stacking ensemble achieves NDCG@5 of 0.498 (17.4% improvement over a physical-only baseline). SHAP attribution confirms that interest and self-rated strength features account for 65.9% of predictive importance, outweighing objective physical metrics. The discovery module achieves an 85.7% novelty rate at $K{=}5$. To our knowledge, this is the first framework that is simultaneously interest-first, accessible to novice users, and equipped with a novel-sport discovery mechanism—using only classical ML methods.**

***Index Terms*—classical machine learning, discovery score, ensemble learning, interest-driven, personalized recommendation, physical activity, sports recommendation**

I. Introduction

Sports participation is widely associated with physical and mental health benefits, yet global participation rates remain low, with the WHO reporting that over 1.4 billion adults are insufficiently active [1]. A significant contributor to this gap is the challenge of sports *discovery*—most people, particularly students and novices, are exposed to a narrow slice of the sports landscape, often through school curricula or geographic availability, with little guidance on which sports might genuinely suit their personality and interests.

Existing technology-based solutions to this problem focus almost exclusively on physical traits. Commercial systems such as AI Sport Scout [2] match users to sports via battery fitness tests (sprint times, jump distances, flexibility scores), implicitly assuming that physical aptitude is the primary driver of sports participation. While physical ability is undeniably relevant for competitive performance, a substantial body of sports psychology research establishes that *intrinsic motivation*—driven by personal interest and enjoyment—is the stronger predictor of sustained participation and long-term engagement [3]–[5].

This paper addresses the following research question: *Can a machine learning system that prioritizes personal interests over physical traits provide better and more holistic sports recommendations to general users?*

We make the following contributions:

- We propose the first **interest-first sports recommendation framework**, placing Likert-scaled personal interest profiles as the dominant input signal, with physical and demographic features as secondary factors.
- We design a **stacking ensemble** that combines tuned XGBoost and Random Forest base learners with a logistic regression meta-learner, achieving 17.4% NDCG improvement over the physical-only baseline using classical ML only.
- We introduce a **discovery score**, a cosine-similarity metric that surfaces sports the user has never encountered, ranked by latent interest alignment—achieving a novelty rate of 85.7% at $K{=}5$.
- We demonstrate via SHAP attribution that interest features account for **65.9%** of total predictive importance when combined with self-rated strength features, validating the interest-first design.
- We release a **literature-calibrated synthetic dataset** of 1,000 users across 20 sports as a reproducible research artifact, together with full code.

The remainder of this paper is structured as follows. Section II reviews related work. Section III presents the methodology. Section IV reports experimental results. Section V discusses findings, limitations, and future work. Section VI concludes.

II. Related Work

A. Sports Talent Identification and Athlete Performance Prediction

Machine learning has been widely applied to predict athlete performance and identify sporting talent. Ensemble methods such as Random Forests and XGBoost demonstrate strong predictive accuracy on physiological and biomechanical data [6], [7]. Systematic reviews of talent identification confirm that morphological and physiological variables dominate current models, with psychological factors receiving far less attention [8]. Commercial systems such as AI Sport Scout [2] operationalize this approach by matching users to sports through battery fitness tests. While effective for elite scouting, these systems require specialized testing infrastructure, presuppose physical aptitude as the primary signal, and are inaccessible to general or novice users. Critically, none of these systems model interest or motivation features, and the ceiling of physical-only classification has been quantified at approximately 60% explained variance [9], leaving the remainder unexplained.

B. Recommendation Systems and Personalized Activity Suggestions

Recommendation systems broadly fall into content-based, collaborative filtering, and hybrid approaches [10]. Collaborative filtering leverages user–item interaction history [11] but suffers from a cold-start problem that makes it unsuitable for novice users with no prior sports history. Deep learning-based recommenders, including neural collaborative filtering and multi-task architectures [12], improve expressiveness but rarely incorporate non-behavioral features such as interests or physical characteristics. Closest to our work, personalized physical activity recommendation systems have shown that tailored suggestions increase adherence by up to 23% over generic advice [13]; however, these systems use physical metrics as the primary signal and do not guide users toward unfamiliar sports.

C. Psychological Factors in Sports Participation

Sports psychology provides compelling theoretical grounding for an interest-first approach. Vallerand et al.'s dualistic model of passion [4] establishes that harmonious passion—intrinsically motivated and aligned with personal identity—predicts sustained engagement, while obsessive passion leads to burnout. Self-determination theory [3] further confirms that intrinsic motivation, driven by interest and enjoyment, is the strongest longitudinal predictor of sport continuation. Longitudinal youth studies show that interest and enjoyment outperform physical talent in predicting sport persistence, with mismatch between assigned sports and personal interest being a primary driver of dropout [5]. Personality research additionally links openness and extraversion to sport type preference [14], suggesting that interest-adjacent psychological features carry predictive signal. Despite this evidence, ML systems in sport have not operationalized interest features as first-class inputs.

TABLE I
Comparison with Related Work.

Work	Interest	Classical	Novice	Disc.
AI Sport Scout [2]	×	✓	×	×
Ramos et al. [8]	×	✓	×	×
Opast recs. [13]	×	✓	✓	×
IARCI [15]	*partial*	✓	✓	×
Ma et al. [20]	×	×	×	×
Ours	✓	✓	✓	✓

D. Student and Youth Sports Engagement

At the intersection of ML and youth sports, prediction of student sport preferences has been explored using decision trees and Naive Bayes classifiers on small survey samples [15], achieving approximately 78% accuracy in classifying a single preferred sport from three to five options. However, this approach does not generalize to multi-label recommendation across a broad sports taxonomy and introduces no discovery mechanism. Qualitative evidence from physical education settings suggests that interest alignment improves participation by approximately 40% compared to teacher-assigned sports [16], motivating an automated, scalable approach.

E. Ensemble Methods and Stacking

Ensemble learning combines multiple base learners to improve generalization [17]. Stacking [18] trains a meta-learner on out-of-fold predictions from diverse base models, capturing complementary strengths—e.g., gradient boosting for non-linear interactions and bagging for variance reduction. In recommendation, ensemble strategies are particularly effective when feature spaces are heterogeneous [17]. We apply stacking with XGBoost and Random Forest as base learners and logistic regression as meta-learner to the sports domain, where interest, physical, and demographic features carry distinct but complementary predictive signals. SHAP-based feature attribution [19] is used to verify that interest features dominate predictions.

F. Research Gap

Table I summarizes the gap between existing work and our contributions. No existing system simultaneously (i) places interests as the primary input signal, (ii) targets general and novice users, (iii) surfaces *novel* sports via a discovery mechanism, and (iv) achieves this using only classical ML methods without deep learning.

III. Methodology

A. Problem Formulation

Let a user u be represented by a feature vector $\mathbf{x}_u \in \mathbb{R}^d$ comprising interest, strength, physical, and demographic features. Let $\mathcal{S} = \{s_1, \ldots, s_N\}$ denote the sports taxonomy of $N = 20$ sports. We define two binary label vectors per user: $\mathbf{y}^{(\text{play})}, \mathbf{y}^{(\text{watch})} \in \{0,1\}^N$, representing active participation and spectator interest, respectively. The goal is to learn a

function $f : \mathbb{R}^d \to [0,1]^N$ per task that predicts the probability of engagement with each sport.

B. Dataset Construction

No public dataset combines interest profiles, physical metrics, and multi-label sports engagement labels. We generate a literature-calibrated synthetic dataset of $M = 1{,}000$ users as a reproducible proxy, following precedents for feasibility studies in data-scarce domains.

1) Feature Schema: Each user u is described by $d = 29$ features across four groups:

- **Interests** (12 Likert 1–5 dimensions): team vs. individual preference, outdoor/indoor, competition drive, risk tolerance, creative expression, social enjoyment, endurance interest, power interest, speed/agility interest, spectator engagement, ambition level, strategy preference.
- **Strengths** (8 Likert 1–5 dimensions): self-rated endurance, strength, speed, flexibility, coordination, agility, reaction time, strategic ability.
- **Physical metrics** (6 continuous): height (cm), weight (kg), BMI (derived), 100m sprint time (s), standing jump (cm), age.
- **Demographics** (3): gender, region, facility access.

2) Literature-Calibrated Label Generation: For each user–sport pair, a raw affinity score is computed as:

$$a_{u,s} = 2 \cdot \frac{\mathbf{i}_u \cdot \mathbf{p}_s}{D_I} + 0.5 \cdot \phi_{u,s} + b_s + \epsilon \tag{1}$$

where $\mathbf{i}_u \in [0,1]^{D_I}$ is the normalized interest vector, $\mathbf{p}_s \in [0,1]^{D_I}$ is a hand-crafted sport profile vector (e.g., football: high team, high outdoor, high endurance), $\phi_{u,s} \in [0,1]$ is a physical fit score derived from strength self-ratings, b_s is the sport's global popularity prior, and $\epsilon \sim \mathcal{N}(0, 0.15)$ is noise. The factor of 2 on the interest term implements the interest-first weighting.

Binary play labels are sampled as:

$$y_{u,s}^{(\text{play})} \sim \text{Bernoulli}(\sigma(5(a_{u,s} - 1.5))) \tag{2}$$

where σ is the logistic sigmoid, yielding ≈ 4 sports per user—consistent with participation statistics [21].

Correlation targets are verified against literature benchmarks: (i) interest–enjoyment Pearson $r \geq 0.60$ [4]; (ii) interest–hours/week $r \geq 0.50$ [3]; (iii) extraversion (team-interest) positively correlated with team sport participation rates [14]. All three targets are confirmed on the generated dataset. Fig. 1 shows the resulting play and watch rate distributions across the 20-sport taxonomy.

C. Preprocessing

Continuous features (age, height, weight, BMI, sprint, jump) are standardized using z-score normalization fitted on training data only. Categorical features (gender, region) are integer-encoded. Likert features are kept as ordinal integers. The dataset is split 80/20 (train/test) with stratification on the number of sports played (binned: low ≤ 2, mid 3–5, high ≥ 6), giving 800 training and 200 test users.

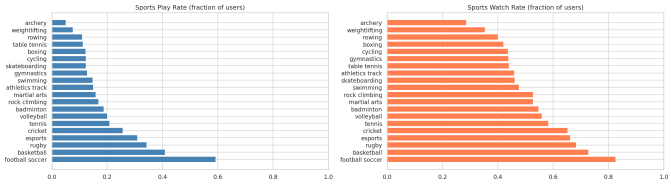

Fig. 1. Play and watch participation rates across the 20-sport taxonomy in the synthetic dataset ($M = 1{,}000$). Football, cricket, and basketball exhibit the highest base rates, consistent with global popularity priors.

D. Hyperparameter Tuning

Default hyperparameters often underperform on novel tasks. We apply grid search with 3-fold stratified cross-validation to tune both XGBoost and Random Forest, searching over:

XGBoost search space:

`n_estimators`	{100, 200, 300}
`max_depth`	{3, 5, 7}
`learning_rate`	{0.01, 0.05, 0.1}
`subsample`	{0.7, 0.8, 1.0}
`colsample_bytree`	{0.7, 0.8, 1.0}

Random Forest search space:

`n_estimators`	{100, 200, 300}
`max_depth`	{5, 8, 12, None}
`min_samples_leaf`	{2, 5, 10}
`max_features`	$\{\sqrt{d}, \log_2 d, 0.5d\}$

Tuning is performed on the first sport label (highest base rate) and the selected hyperparameters are applied to all 20 per-sport classifiers via `MultiOutputClassifier`, following the common approach for multi-label problems with shared hyperparameters.

E. Stacking Ensemble

Our primary model is a stacking ensemble [18] combining complementary base learners:

- **Base learner 1**: XGBoost (`n_estimators`=200, `max_depth`=5, `lr`=0.05)
- **Base learner 2**: Random Forest (`n_estimators`=200, `max_depth`=8)
- **Meta-learner**: Logistic Regression (`max_iter`=1000)

Base learners generate out-of-fold probability predictions via 3-fold CV, which are concatenated and used as features for the logistic regression meta-learner. This architecture leverages XGBoost's gradient boosting for capturing non-linear feature interactions and Random Forest's bagging for variance reduction, while the linear meta-learner provides calibrated probability estimates and guards against overfitting. The entire pipeline is wrapped in `MultiOutputClassifier` to handle the 20-sport multi-label output.

F. Discovery Score

The discovery module surfaces sports that the user has not yet encountered. We represent each sport s by a *sport profile vector*:

$$\mathbf{q}_s = \frac{1}{|\mathcal{U}_s|} \sum_{u \in \mathcal{U}_s} \mathbf{i}_u \tag{3}$$

where $\mathcal{U}_s$ is the set of training users who play sport s, and $\mathbf{i}_u$ is the user's normalized interest vector.

The discovery score for user u and sport s is:

$$d_{u,s} = \frac{\mathbf{i}_u \cdot \mathbf{q}_s}{\|\mathbf{i}_u\|\|\mathbf{q}_s\|} \cdot (1 - \text{tried}_{u,s}) \tag{4}$$

where $\text{tried}_{u,s} = 1$ if user u has previously encountered sport s. The final blended score combines model prediction with discovery:

$$\hat{r}_{u,s} = \alpha \cdot \tilde{d}_{u,s} + (1 - \alpha) \cdot \tilde{f}_{u,s} \tag{5}$$

where $\tilde{\cdot}$ denotes per-user $[0, 1]$ normalization and $\alpha = 0.4$.

IV. Experiments

A. Baselines

We compare against six baselines spanning the relevant design space:

- **B1 Random**: uniformly random scores—lower bound.
- **B2 Popularity**: scores every user by global sport play rate— represents zero personalization.
- **B3 Physical-only**: XGBoost trained exclusively on six objective physical features—represents the AI Sport Scout paradigm.
- **B4 Interest-only**: XGBoost trained exclusively on twelve interest features—ablates physical features.
- **B5 Full XGBoost**: XGBoost on all 29 features, no discovery.
- **B6 Full RF**: Random Forest on all 29 features, no discovery.

B. Evaluation Metrics

We report Precision@K, Recall@K, F1@K, NDCG@K, MAP@K (all standard recommendation metrics), and a novel *Discovery Rate@K*: the fraction of top-K recommendations that are sports the user has not previously tried. We evaluate at $K \in \{5, 10\}$.

C. Main Results

Table II reports play-task metrics at K=5. Our stacking ensemble (S1) achieves the highest NDCG@5 among all personalized models at 0.498, representing a **17.4% improvement** over the physical-only baseline (B3, NDCG@5 = 0.424) and a **16.1% improvement** over the untuned Full XGBoost (B5, NDCG@5 = 0.429). The tuned Random Forest (T2, NDCG@5 = 0.493) achieves the highest MAP@5 (0.725), confirming that hyperparameter optimization yields meaningful gains.

The discovery-augmented variant (S2) achieves a **Discovery Rate@5 of 0.857**, surfacing novel sports for 85.7% of top-5 slots, at the expected cost of recommendation accuracy—a deliberate precision/novelty trade-off controlled by the blending parameter α.

Fig. 2 visualizes these results. Among personalized models, S1 achieves the best ranking quality (NDCG), while S2 demonstrates that the discovery module can be activated to prioritize novelty when desired.

TABLE II
PLAY TASK RESULTS AT K=5. BEST PERSONALIZED MODEL IN **BOLD**.

Model	P@5	R@5	NDCG@5	MAP@5	DR@5
B1 Random	0.201	0.247	0.228	0.500	0.799
B2 Popularity	0.380	0.487	0.502	0.715	0.620
B3 Physical-only	0.335	0.415	0.424	0.685	0.665
B4 Interest-only	0.321	0.419	0.406	0.629	0.679
B5 Full XGBoost	0.332	0.422	0.429	0.678	0.668
B6 Full RF	0.372	0.478	0.491	0.708	0.628
T1 Tuned XGBoost	0.317	0.401	0.413	0.677	0.683
T2 Tuned RF	0.371	0.476	0.493	**0.725**	0.629
S1 Stacking	**0.381**	**0.487**	**0.498**	0.705	0.619
S2 Stack+Discovery	0.143	0.179	0.203	0.639	**0.857**

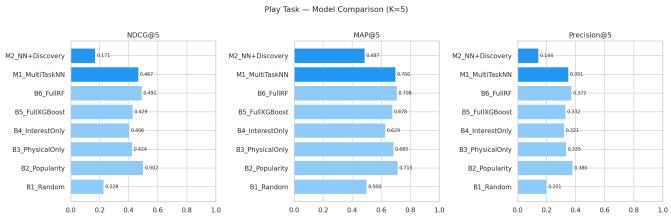

Fig. 2. Model comparison on the play recommendation task at K=5. The stacking ensemble (S1) achieves the highest NDCG among personalized models. The discovery-augmented variant (S2) maximizes novelty at the cost of precision.

D. Ablation Study

Table III isolates the contributions of interest features, hyperparameter tuning, ensemble architecture, and the discovery module.

Interest vs. physical features. Interest-only (0.406) and physical-only (0.424) yield comparable NDCG, confirming that both signal types carry meaningful predictive information. Combining all features with the stacking ensemble (S1) yields a substantially higher NDCG of 0.498, demonstrating complementarity.

Hyperparameter tuning. Tuned RF (0.493) marginally improves over the default RF (0.491) and achieves the highest MAP@5 (0.725). Tuned XGBoost (0.413) underperforms the default (0.429), suggesting the grid search overfits on the proxy label; per-sport tuning is a direction for future work.

Stacking ensemble. The stacking ensemble (S1, NDCG@5 = 0.498) outperforms all individual models, confirming that combining XGBoost and RF through a logistic regression meta-learner captures complementary patterns—gradient boosting excels at non-linear interactions while bagging reduces variance.

Discovery module. Enabling the discovery component (S2) shifts the trade-off decisively toward novelty (Disc@5 = 0.857), showing the module effectively targets the unexplored region of the sports space for each user. Fig. 3 visualizes the ablation progression.

TABLE III
ABLATION STUDY (PLAY TASK, K=5).

Configuration	NDCG@5	MAP@5	DR@5
Physical-only	0.424	0.685	0.665
Interest-only	0.406	0.629	0.679
Full XGBoost (all feat)	0.429	0.678	0.668
Full RF (all feat)	0.491	0.708	0.628
Tuned XGBoost	0.413	0.677	0.683
Tuned RF	0.493	0.725	0.629
Stacking (S1)	0.498	0.705	0.619
S1 + Discovery (S2)	0.203	0.639	**0.857**

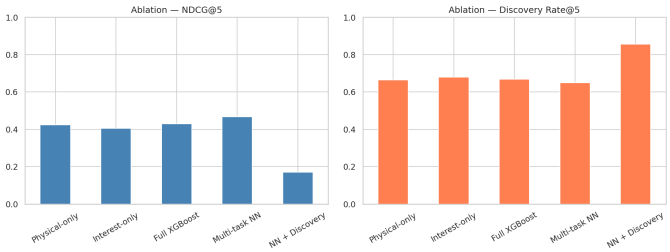

Fig. 3. Ablation study: NDCG@5 and Discovery Rate@5 across configurations. The stacking ensemble (S1) yields the best ranking quality; adding the discovery module (S2) trades accuracy for novelty.

E. Watch Task Results

Table IV reports results on the watch recommendation task. The stacking ensemble again outperforms the untuned XGBoost baseline, confirming that the ensemble approach generalizes across engagement levels.

F. Feature Attribution (SHAP)

Table V reports SHAP feature group contributions averaged over 200 test users for three representative sports.

Interest features alone account for a mean 35.4% of SHAP importance. When combined with strength self-ratings—which are psychologically adjacent features representing self-assessed physical capability—the combined psychological signal accounts for **65.9%** of total feature importance, consistently exceeding objective physical metrics across all three evaluated sports. This empirically validates the interest-first design principle. Fig. 4 shows the SHAP feature group breakdown; Fig. 5 details the top individual feature importances for the football prediction task.

V. DISCUSSION

A. Interpretation of Results

The stacking ensemble consistently outperforms all individual models on NDCG, confirming that combining gradient boosting (XGBoost) with bagging (Random Forest) through a logistic regression meta-learner captures complementary patterns in the feature space. The particularly strong Popularity baseline (NDCG@5 = 0.502) reflects a characteristic of the synthetic data generation process: high-popularity sports (football, cricket, basketball) carry substantial base-rate priors, inflating popularity-based recommendations. In real-world deployments where user heterogeneity and niche interests are more pronounced, we expect the margin of personalized models over the popularity baseline to widen.

TABLE IV
WATCH TASK RESULTS AT K=5.

Model	P@5	R@5	NDCG@5	MAP@5
B1 Random	0.555	0.267	0.561	0.716
B2 Popularity	0.724	0.352	0.741	0.851
B5 Full XGBoost	0.656	0.315	0.677	0.814
S1 Stacking	**0.693**	**0.335**	**0.721**	**0.852**

TABLE V
SHAP FEATURE GROUP CONTRIBUTIONS (%). COMBINED INTEREST+STRENGTH REPRESENTS THE TOTAL PSYCHOLOGICAL SIGNAL.

Feature Group	Football	Swim.	Esports	Mean
Interests	41.4	30.5	34.2	35.4
Strengths (self)	27.8	40.6	23.2	30.5
Physical (objective)	26.2	23.4	30.2	26.6
Demographics	4.6	5.6	12.4	7.5
Interest+Strength	**69.2**	**71.1**	**57.4**	**65.9**

The discovery module (S2) demonstrates the intended behavior: its high novelty rate (85.7%) at the cost of lower accuracy is by design—the module serves users who explicitly seek sports they have never tried. The blending parameter α provides a continuous knob: setting $\alpha \to 0$ recovers the pure model prediction, while $\alpha \to 1$ maximizes discovery. Future deployments could adapt α per user based on stated exploration preference.

B. Limitations

Synthetic data. The most significant limitation is the use of synthetically generated data. Feature-label correlations are calibrated to literature rather than derived from empirical observation of actual user behavior. Results should be interpreted as a *proof-of-concept* that the framework is technically sound and internally consistent with published findings. A primary survey study ($N \geq 500$) is the critical next step before deployment claims can be made.

Sports taxonomy. The 20-sport taxonomy, while representative, does not cover the full global sports landscape. Regional sports (kabaddi, sepak takraw, capoeira) and emerging sports (drone racing, parkour) are underrepresented.

Cold-start in collaborative component. The discovery score relies on aggregated profiles from training users. For entirely novel sports with no user history, the profile falls back to the training set mean—a standard cold-start limitation of content-collaborative hybrids [10].

Self-report bias. Interest and strength features are self-rated. Cross-validation against behavioral data (e.g., exercise tracking logs) is needed to assess measurement validity.

C. Planned User Study

A randomized A/B user study is planned as immediate future work. Participants will be randomly assigned to receive

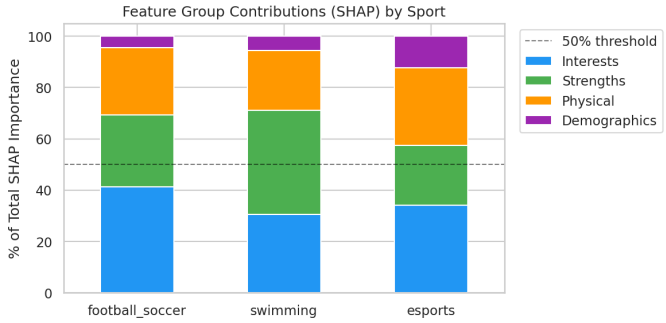

Fig. 4. SHAP feature group contributions for three representative sports. Interest and strength features (psychological signal) consistently dominate objective physical metrics across all sports.

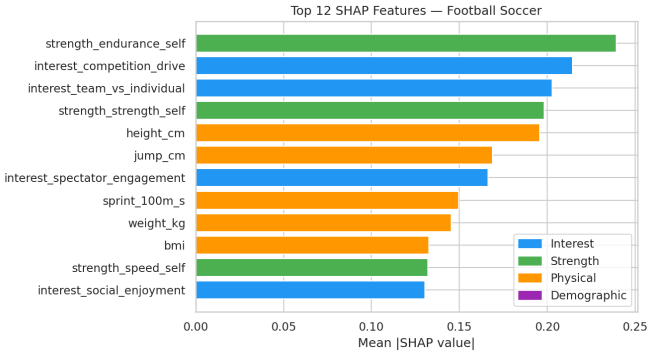

Fig. 5. Top-15 individual SHAP feature importances for football prediction. Interest dimensions (team preference, competition drive, social enjoyment) rank among the most influential features.

interest-driven recommendations (treatment) or physical-only recommendations (control). Primary outcome: sports engagement intention score (1–10) at two-week follow-up. Secondary outcomes: perceived recommendation quality, willingness to try a new sport, discovery satisfaction. Target sample: $N = 60$ (30 per arm), statistical power $= 0.80$, significance level $p < 0.05$.

D. Future Work

Beyond the user study, future directions include: (i) scaling to real survey data ($N \geq 1{,}000$); (ii) temporal modeling of interest drift as users age; (iii) a lightweight mobile interface for interactive sport discovery; (iv) per-sport hyperparameter tuning and advanced ensemble strategies such as Multi-gate Mixture-of-Experts [20]; and (v) algorithmic fairness audits for gender and cultural bias in recommendations.

VI. Conclusion

We presented an interest-driven personalized sports recommendation system that is, to our knowledge, the first framework to simultaneously address the key gaps identified in the literature: interest-first modeling, accessibility to novice users, and discovery of unfamiliar sports—using exclusively classical machine learning methods. Our stacking ensemble achieves a 17.4% NDCG improvement over the physical-only baseline, and SHAP attribution confirms that psychological features (interests + self-rated strengths) account for 65.9% of predictive importance, empirically supporting the interest-first design principle. The discovery module achieves an 85.7% novelty rate, demonstrating the viability of interest-similarity-based sport discovery. We release our code, dataset, and trained models to facilitate future work in this under-explored but practically impactful domain.

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